

8.4.2

$$E = -\frac{1}{c} \frac{\partial A}{\partial t} - \nabla \phi$$

$$B = \nabla \times A$$

now, take

$$\begin{cases} \phi \rightarrow \phi' = \phi + \frac{1}{c} \frac{\partial \Lambda}{\partial t} \\ A \rightarrow A' = A - \nabla \Lambda \end{cases}$$

$$\Rightarrow E' = -\frac{1}{c} \frac{\partial A'}{\partial t} - \nabla \phi'$$

$$= -\frac{1}{c} \frac{\partial A}{\partial t} + \frac{1}{c} \frac{\partial \nabla \Lambda}{\partial t} - \nabla \phi - \frac{1}{c} \nabla \frac{\partial \Lambda}{\partial t}$$

$$= -\frac{1}{c} \frac{\partial A}{\partial t} - \nabla \phi$$

and $B' = \nabla \times A'$

$$= \nabla \times (A - \nabla \Lambda)$$

$$= \nabla \times A - \underbrace{(\nabla \times \nabla) \Lambda}_{0 \text{ by vector identity}}$$

$$= \nabla \times A$$

So the fields computed using $(\phi, \vec{A})$ and $(\phi', \vec{A}')$ are identical.

$$\begin{aligned}
 (1) \quad \phi' &= \phi + \frac{1}{c} \frac{\partial \Lambda}{\partial t} \\
 &= \phi + \frac{1}{c} \partial_t \left(-c \int_{-\infty}^t \phi(r, t') dt' \right) \quad \left(\text{recall: } \partial_x \int_a^x f(x') dx' = f(x) \right) \\
 &= \phi - (\phi(r, t)) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 A' &= A - \nabla \Lambda = A + c \int_{-\infty}^t \nabla \phi(r, t') dt' \\
 \nabla \cdot A' &= \nabla \cdot A + c \int_{-\infty}^t \nabla^2 \phi(r, t') dt'
 \end{aligned}$$

$\neq 0$ in general.

5.1 If we had left in the $e^{i\vec{k}\cdot\vec{r}}$, Eqn 18.5.19 would read

$$H_{fi}^1 = N \int e^{-i\vec{p}_f \cdot \vec{r}/\hbar} A_0 \cdot (-i\hbar \nabla) e^{-r/a_0} d^3r e^{i\vec{k} \cdot \vec{r}}$$

$$= N \int e^{-i(\vec{p}_f - \hbar\vec{k}) \cdot \vec{r}/\hbar} A_0 \cdot (-i\hbar \nabla) e^{-r/a_0} d^3r$$

Integrating by parts, 18.5.21 becomes

$$H_{fi}^1 = N A_0 \cdot (\vec{p}_f - \hbar\vec{k}) \int e^{-i(\vec{p}_f - \hbar\vec{k}) \cdot \vec{r}/\hbar} e^{-r/a_0} d^3r \quad (\text{choose } \hat{z} \text{ along } \vec{p}_f)$$

$$= \frac{8\pi/a_0}{\left(\frac{1}{a_0}\right)^2 + \frac{|\vec{p}_f - \hbar\vec{k}|^2}{\hbar^2}} \quad (18.5.22)$$

$$\Rightarrow R_{i \rightarrow f} = \frac{2\pi}{\hbar} \left(\frac{e}{2mc}\right)^2 \frac{1}{8\pi^3 \hbar^3} \frac{1}{\pi a_0^3} \frac{|\vec{A}_0 \cdot (\vec{p}_f - \hbar\vec{k})|^2}{\left[1 + \frac{a_0^2 |\vec{p}_f - \hbar\vec{k}|^2}{\hbar^2}\right]^4} \delta(E_f^0 - E_i^0 - \hbar\omega) \quad (18.5.23)$$

This is the average transition rate after a long exposure to coherent light into a precise free particle state $|p_f\rangle$. Realistically, we are more interested in the rate of production of plane wave states pointed in directions lying in a cone of solid angle $d\Omega$, with any allowed energy $E_f = \frac{p_f^2}{2m}$.

$$\Rightarrow \text{compute } d\Omega \int dp_f p_f^2 R_{i \rightarrow f}. \quad \text{Now } \int \delta(f(x)) g(x) dx = \int \delta(y) g(f^{-1}(y)) \frac{df^{-1}(y)}{dy} dy$$

$$= g(f^{-1}(0)) \frac{df^{-1}(y)}{dy} \Big|_{y=0}$$

$$\Rightarrow \text{for } f(x) = \frac{p_f^2}{2m} - E_i^0 - \hbar\omega, \quad \frac{df^{-1}(y)}{dy} = \frac{m}{p_f} \Rightarrow g(p_f) \frac{m}{p_f} \Big|_{p_f = \sqrt{2m(E_i^0 + \hbar\omega)}}$$

$$\Rightarrow R_{i \rightarrow d\Omega} = \frac{4e^2 a_0^3 p_f |\vec{A}_0 \cdot (\vec{p}_f - \hbar\vec{k})|^2 d\Omega}{m\pi\hbar^4 c^2 \left[1 + \frac{a_0^2 |\vec{p}_f - \hbar\vec{k}|^2}{\hbar^2}\right]^4} \Big|_{|\vec{p}_f| = \sqrt{2m(E_i^0 + \hbar\omega)}}$$

but we worked in the Coulomb gauge, where $\vec{A} \cdot \vec{k} = 0$

$$\therefore R_{i \rightarrow d\Omega} = \frac{4e^2 a_0^3 p_f |\vec{A}_0 \cdot \vec{p}_f|^2 d\Omega}{m\pi\hbar^4 c^2 \left[1 + \frac{a_0^2 |\vec{p}_f - \hbar\vec{k}|^2}{\hbar^2}\right]^4} \Big|_{|\vec{p}_f| = \sqrt{2m(E_i^0 + \hbar\omega)}}$$

This is maximized when $\vec{p}_f$ and $\vec{k}$ are parallel, reflecting the fact that the electron has an increased component in the high energy plane wave state after the perturbation if the initial e^- momentum and the photon momentum could add constructively.

$$(1) \quad \sigma = \frac{128 a_0^3 \pi e^2 P_f^2}{3 m \hbar^3 \omega c \left[1 + \frac{P_f^2 a_0^2}{\hbar^2} \right]^4} \quad E_f = \frac{P_f^2}{2m} = 10 R_y = \frac{10 m e^4}{2 \hbar^2}$$

$$\text{Now } a_0 = \frac{\hbar^2}{m e^2}, \text{ so } \frac{P_f^2 a_0^2}{\hbar^2} = \frac{2m \cdot 10 m e^4 \hbar^4}{\hbar^2 \cdot 2 \hbar^2 m^2 e^4} = 10$$

$$\Rightarrow \sigma = \frac{128 \pi e^2}{3 m \omega c} \cdot 10^{3/2} \cdot \frac{1}{(1+10)^4}$$

And the e^- started in the hydrogen ground state, $E_i = -R_y$

$$\Rightarrow 10 R_y = -R_y + \hbar \omega$$

$$\Rightarrow \hbar \omega = 11 R_y = \frac{11 m e^4}{2 \hbar^2}$$

$$\text{So } \sigma = \frac{128 \pi \cdot 10^{3/2}}{3 \cdot 11^4} \cdot \frac{e^2 \hbar}{m c} \cdot \frac{2 \hbar^2}{11 m e^4}$$

$$\frac{2 \hbar^3}{11 m^2 c e^2} = \frac{2 \hbar^4 e^2}{11 m^2 c e^4 \hbar} = \frac{2}{11} a_0^2 \frac{e^2}{\hbar c} = \frac{2}{11} \alpha a_0^2$$

$$\Rightarrow \sigma = \frac{128 \cdot 10^{3/2}}{3 \cdot 11^4} \cdot \frac{2}{11} \alpha \cdot \pi a_0^2, \quad \alpha = \frac{1}{137}, \quad \sqrt{10} \approx 3$$

$$\approx \frac{(128/\alpha)}{1} \cdot \frac{(10^{3/2}/3)}{1} \cdot \frac{(2/\alpha)}{1} \cdot \frac{(1/\sqrt{4})}{1} = \boxed{(\pi a_0^2) \cdot 10^{-4}}$$

(2) In the limit $\frac{P_f a_0}{\hbar^2} \gg 1$, $\sigma \rightarrow \frac{128 \pi e^2 \hbar^5}{P_f^5 a_0^5 \cdot 3 m \omega c}$. The Z could appear either

with e^2 or a_0 . But the e^2 came from $(H_{fi})^2$, the interaction between light & the electron - this doesn't care about the nucleus charge. On the other hand, we know the Bohr radius is tightened to a_0/Z . So $\sigma \sim Z^5$, because $\sigma \sim a_0^{-5}$.